Exhibits (tables and figures) for: Revealing hidden jump strategies beyond conventional force-plate metrics

Tables

Table 1. Quality-control flow (main analysis).

Step	Trials	Sessions	Athletes
All male football CMJ trials	833	333	215
R1: valid body mass	828	329	212
R2: quiet-stance mass within 5% of body mass	813	328	212
R3: no force on plate during flight	811	327	212
R4: complete metrics and age	811	327	212
R5: end of landing (lowest COM) found	805	327	212

Note. Rules R1–R4 apply to both landing definitions; R5 applies only to the main landing definition (touch-down to the lowest point of the centre of mass), and the 1-s landing window therefore retained 811 trials. Sessions are test occasions on which an athlete performed one or more trials. COM = centre of mass.

Table 2. Characteristics of the analysed sample (805 trials, 212 athletes).

Variable	*n*	Mean ± SD	Median (IQR)	Range
Age at first test (years)	212	16.01 ± 5.94	14.10 (12.00–16.80)	8.80–37.90
Body mass at first test (kg)	212	53.31 ± 15.90	51.73 (40.86–64.03)	25.03–95.97
Jump height, flight time (cm)	805	28.19 ± 8.43	26.96 (21.62–34.69)	8.54–51.47
Concentric peak force (N)	805	1269.26 ± 420.74	1227.01 (923.47–1582.99)	473.52–3073.22
Concentric peak force (× BW)	805	2.35 ± 0.38	2.29 (2.11–2.51)	1.65–4.70
Eccentric braking impulse (N s)	805	27.80 ± 15.43	24.60 (16.59–35.70)	0.00–95.18
Concentric impulse, first 100 ms (N s)	805	62.24 ± 26.66	58.70 (41.76–79.14)	1.51–187.93
Peak landing force (× BW)	805	5.39 ± 1.97	4.97 (3.88–6.67)	2.19–14.88
Contraction time (s)	805	0.78 ± 0.19	0.77 (0.66–0.90)	0.31–1.48
Countermovement depth (cm)	805	−26.55 ± 9.90	−27.13 (−33.55–−20.71)	−52.39–0.00
Touch-down–lowest COM (s)	805	0.25 ± 0.11	0.24 (0.15–0.31)	0.07–0.93
Trials per athlete	212	3.80 ± 2.28	3.00 (2.00–5.00)	1.00–16.00
Trials per session	327	2.46 ± 0.78	2.00 (2.00–3.00)	1.00–7.00
Sessions per athlete	212	1.54 ± 0.80	1.00 (1.00–2.00)	1.00–6.00

Note. Age and body mass are per athlete at the first analysed trial; all other rows are per trial unless stated. Jump height is derived from flight time. Countermovement depth is the vendor estimate of centre-of-mass displacement (negative downwards); one trial with zero depth and zero braking impulse satisfied all quality-control rules and was retained. Touch-down to lowest COM is the duration of the landing phase in the main analysis. BW = body weight; COM = centre of mass; IQR = interquartile range.

Table 3. Retained principal components of the concatenated take-off and landing curve (main analysis).

Component	Variance (%)	Cumulative (%)	Landing share of loading (%)	Largest loading at	Pearson r with LPF	JH	CT	CMD	LD
PC1	37.3	37.3	99.7	Landing 40%	0.39	0.13	−0.04	0.01	−0.75
PC2	16.0	53.3	99.9	Landing 26%	0.22	0.13	0.02	0.00	−0.14
PC3	12.6	65.9	99.5	Landing 16%	0.39	0.34	0.00	−0.14	−0.32
PC4	8.9	74.8	99.3	Landing 9%	0.45	0.24	0.15	−0.18	−0.04

Note. Landing share of loading is the percentage of the squared loading vector in the landing block (the remainder is in the take-off block). Largest loading at gives the phase and the percentage of the phase at which the loading has its largest absolute value. LPF = peak landing force ($\times$ BW); JH = jump height; CT = contraction time; CMD = countermovement depth; LD = landing duration (touch-down to lowest COM). Signs are fixed so that the largest absolute loading is positive.

Table 4. Dispersion of component scores in the full sample and among trials matched on jump height or on relative concentric peak force (main analysis).

Component	SD, full sample	Jump-height band		Relative peak-force band	
		SD	Ratio (95% CI)	SD	Ratio (95% CI)
PC1	7.28	6.45	0.89 (0.70–1.06)	7.33	1.01 (0.77–1.16)
PC2	4.77	4.62	0.97 (0.77–1.16)	5.01	1.05 (0.82–1.24)
PC3	4.22	4.03	0.95 (0.74–1.13)	4.25	1.00 (0.81–1.17)
PC4	3.56	4.16	1.17 (0.81–1.44)	3.75	1.05 (0.87–1.23)
Whole curve[a]	11.92	11.99	1.01	11.95	1.00

Note. Matched bands comprise trials between the 45th and 55th percentiles of jump height (26.0–28.2 cm; 82 trials, 51 athletes) or of concentric peak force relative to body weight (2.25–2.32 BW; 81 trials, 58 athletes). Ratio = SD in the band divided by SD in the full sample; 95% CI from 2,000 athlete-level bootstrap resamples, conditional on the estimated components. Component SDs depend on the number of grid points and are comparable only within this table; ratios are scale-free. [a]Square root of the summed point-wise variance of the 202-point curve.

Table 5. Variance of each component score explained by the four vendor metrics, age and body mass, under four evaluation schemes (main analysis, R^2 in %).

Component	In-sample	Adjusted	Athlete CV	One trial per athlete	Athlete means
PC1	7.7	7.0	4.0	10.0	11.1
PC2	2.7	2.0	0.3	4.3	5.4
PC3	13.1	12.4	11.8	17.8	21.2
PC4	9.1	8.4	6.3	11.1	13.8

Note. Predictors: jump height, concentric peak force, eccentric braking impulse, concentric impulse over the first 100 ms, age and body mass (ordinary least squares). In-sample and adjusted values use all 805 trials. Athlete CV = out-of-sample R^2 from five-fold cross-validation with athletes allocated to folds. One trial per athlete = one randomly drawn trial per athlete (212 trials). Athlete means = scores and predictors averaged within athlete.

Table 6. Decomposition of component-score variance between athletes, sessions and trials (main analysis, % of total variance).

Component	Between athletes	Between sessions	Within session	Between athletes, age-adjusted
PC1	29.6 (13–47)	34.9 (16–53)	35.5 (26–46)	28.2
PC2	21.4 (4–37)	17.0 (3–36)	61.6 (50–73)	20.6
PC3	22.9 (12–31)	2.3 (0–14)	74.8 (63–85)	15.9
PC4	10.0 (0–23)	20.3 (9–32)	69.6 (59–79)	7.1

Note. Three-level random-intercept model (athlete, session within athlete, trial within session) fitted by restricted maximum likelihood. Values in parentheses are 95% CIs from 2,000 athlete-level bootstrap resamples. Only 83 of the 212 athletes contributed more than one session, so the between-session share is imprecise. Age-adjusted = the same model with age at test as a fixed effect.

Table 7. Main analysis compared with the sensitivity analysis in which the landing phase ended 1 s after touch-down.

Quantity	Main: landing to lowest COM	Sensitivity: 1-s landing
Trials / athletes after QC	805 / 212	811 / 212
Variance explained, PC1–PC4 (%)	37.3, 16.0, 12.6, 8.9	20.2, 16.6, 15.1, 10.2
Cumulative, PC1–PC4 (%)	74.8	62.2
Landing share of loading, PC1–PC4 (%)	99.7, 99.9, 99.5, 99.3	96.4, 92.5, 3.7, 80.2
PC1 SD ratio, jump-height band (95% CI)	0.89 (0.70–1.06)	0.82 (0.69–0.96)
PC3 SD ratio, relative-peak-force band (95% CI)	1.00 (0.81–1.17)	0.74 (0.57–0.87)
Whole-curve SD ratio, jump-height band	1.01	1.01
In-sample R^2, PC1–PC4 (%)	7.7, 2.7, 13.1, 9.1	19.1, 5.3, 68.3, 17.8
Athlete-CV R^2, PC1–PC4 (%)	4.0, 0.3, 11.8, 6.3	13.4, 2.8, 66.4, 15.1
Between-athlete variance, PC1–PC4 (%)	29.6, 21.4, 22.9, 10.0	45.3, 18.7, 26.5, 31.6
r(PC1, landing peak force)	0.39	0.71
r(PC3, contraction time)	0.00	−0.79

Note. COM = centre of mass; CI = confidence interval; CV = cross-validation. Between-athlete variance is taken from the three-level random-intercept model. In the 1-s window PC3 is a take-off component (Fig. 5), which explains its correlation with contraction time and its narrower dispersion among trials matched on relative peak force.

Figures

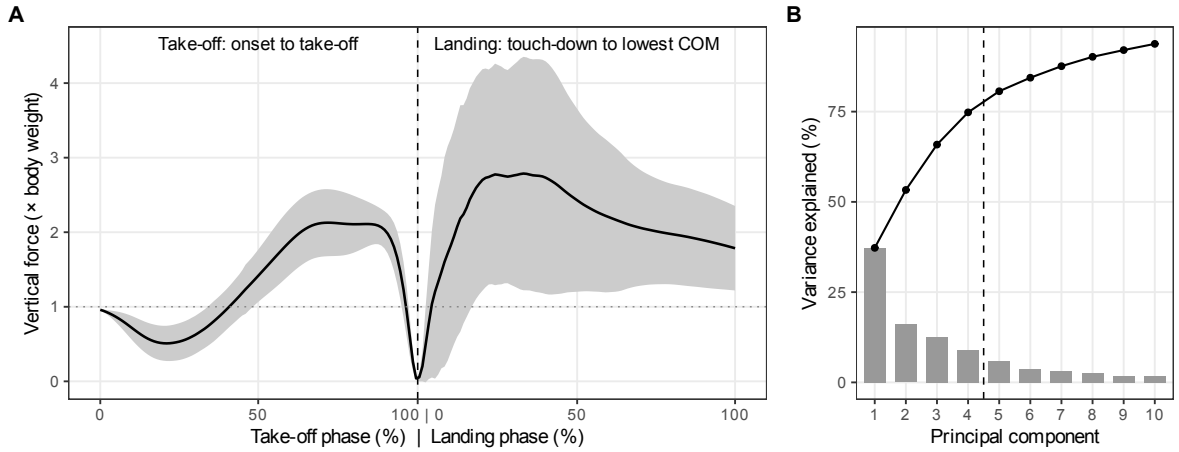


Fig. 1. (A) Mean body-weight-normalised vertical force (line) ± 1 SD (shaded) of the 805 analysed countermovement jumps. The take-off phase (movement onset to take-off) and the landing phase (touch-down to the lowest point of the centre of mass, COM) were each time-normalised to 0–100% and concatenated; flight is omitted. The dotted line marks body weight. (B) Variance explained by each of the first ten principal components (bars) and cumulatively (line); the dashed line marks the four retained components.

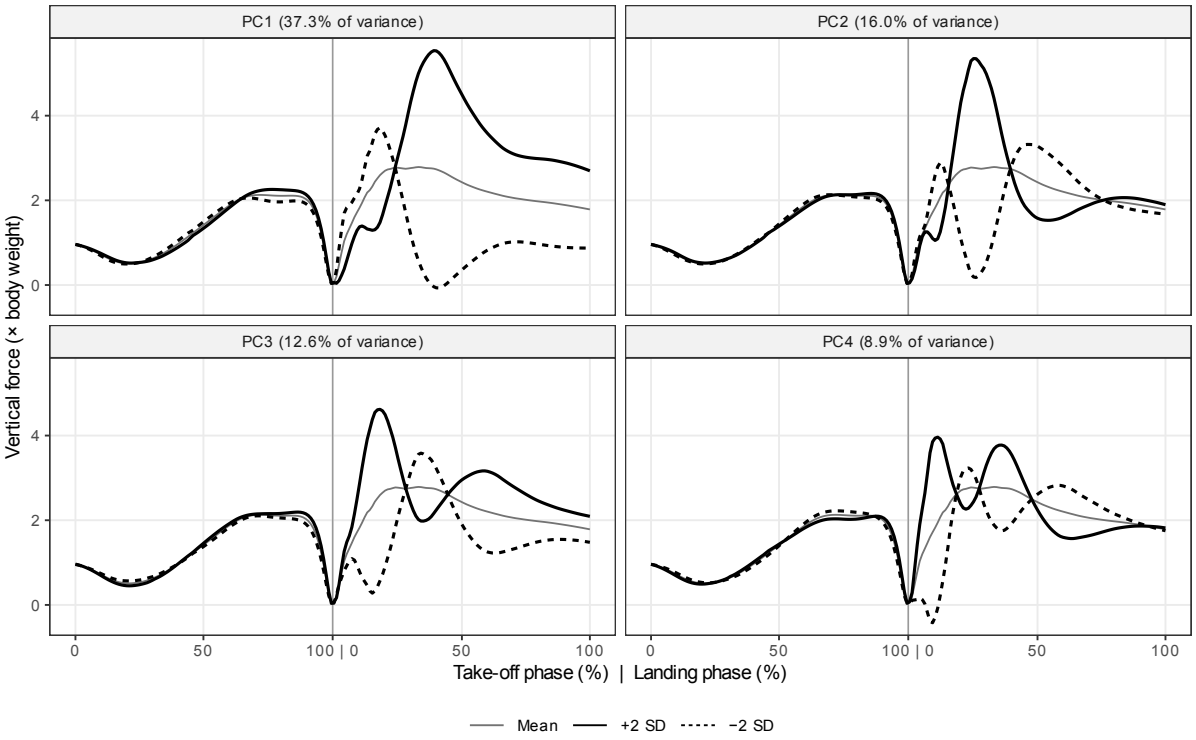


Fig. 2. Interpretation of the four retained principal components (main analysis). Each panel shows the mean curve (thin grey line) and the mean curve plus (solid black line) or minus (dashed black line) two standard deviations of the component score. Values below zero in the reconstructions are an artefact of adding a linear component to the mean and do not correspond to observed trials. The vertical line separates the take-off and landing phases.

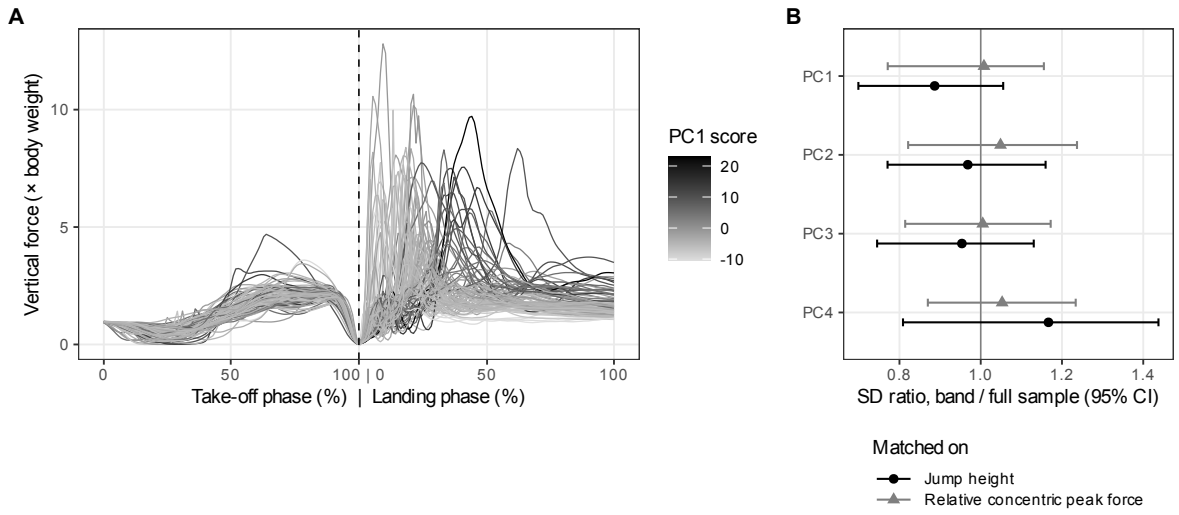


Fig. 3. Trials matched on jump height do not share a common force–time curve. (A) Curves of the 82 trials between the 45th and 55th percentiles of jump height (27.1 ± 0.65 cm), shaded from light grey (low) to black (high) by their PC1 score. (B) Ratio of the standard deviation of each component score among trials matched on jump height (black circles) or on relative concentric peak force (grey triangles) to that in the full sample, with athlete-level bootstrap 95% confidence intervals. A ratio of 1 indicates that matching on the metric leaves curve-shape variation unchanged.

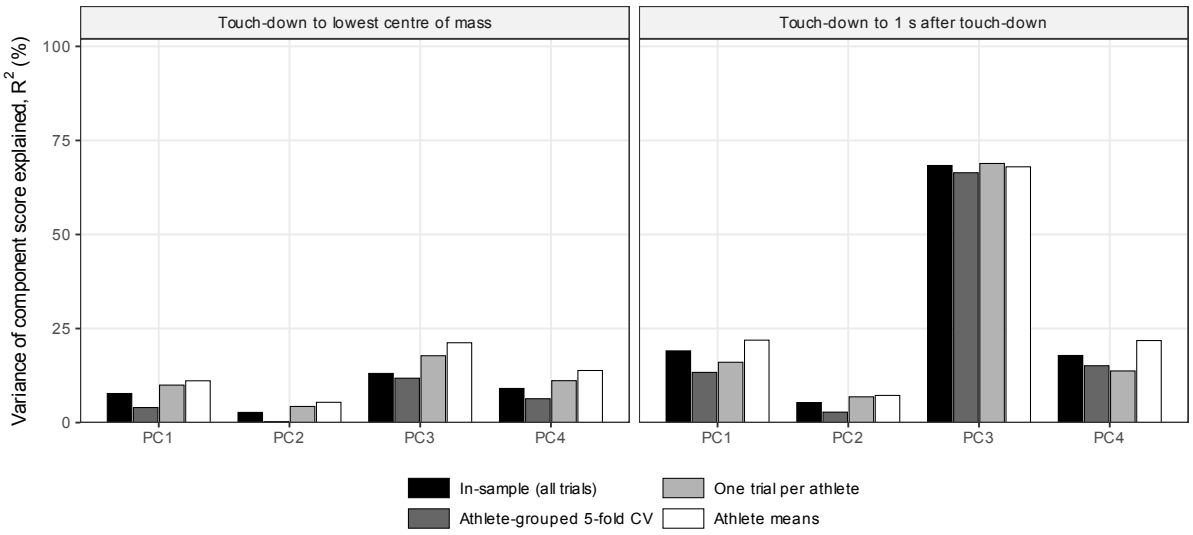


Fig. 4. Proportion of the variance of each component score explained by the four vendor metrics, age and body mass, under four evaluation schemes (bars from black to white: in-sample, athlete-grouped cross-validation, one trial per athlete, athlete means), for the main landing definition (left) and the 1-s landing window (right). CV, cross-validation with athletes allocated to folds.

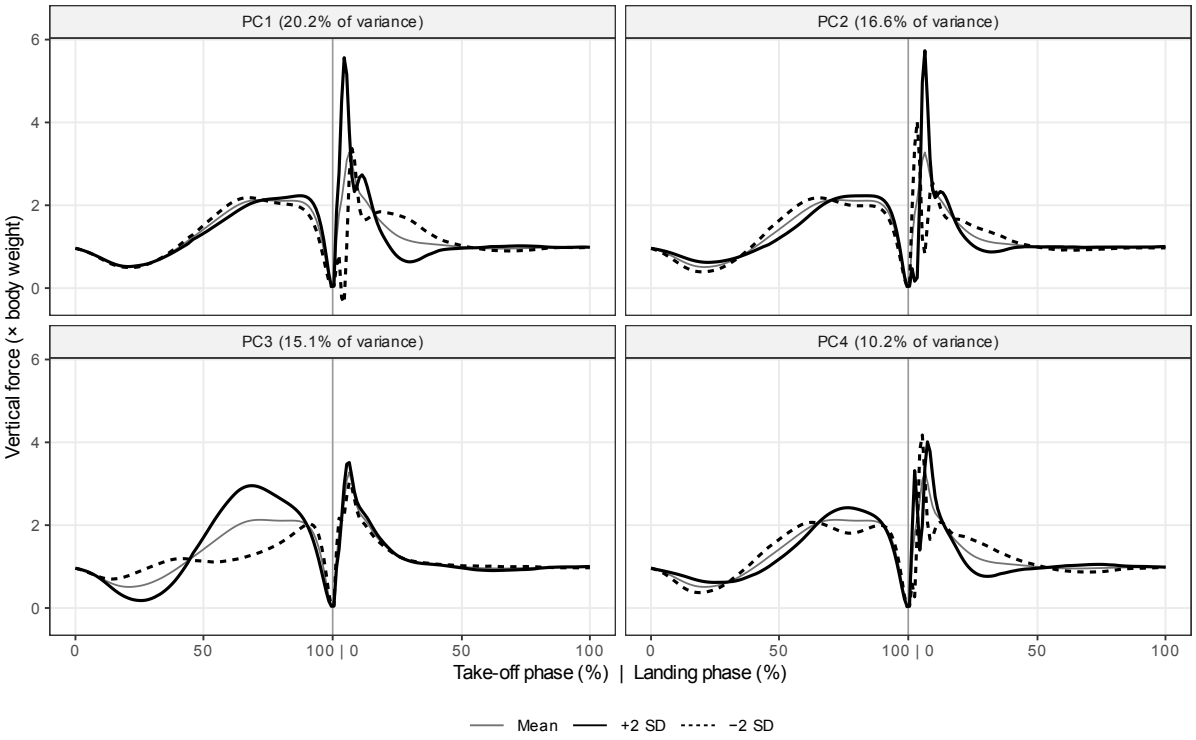


Fig. 5. Sensitivity analysis with the landing phase ending 1 s after touch-down: mean curve (thin grey line) and the mean curve plus (solid black line) or minus (dashed black line) two standard deviations of each retained component. Under this definition PC3 is located almost entirely in the take-off phase (96.3% of its squared loading) and describes the duration and amplitude of the propulsive force.